

Ahmed Abd Elhamid Salem

Senior Full-Stack Developer | .NET and Angular

Nasr City, Cairo, Egypt | +20 101 581 9700

ahmedsalem1041998@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 5+ years of experience designing, developing, and deploying production-grade web applications using .NET 8, ASP.NET Core, C#, and Angular (versions 17 to 21). Proven expertise in Clean Architecture, Command Query Responsibility Segregation (CQRS), RESTful API design, real-time systems with SignalR and Redis, payment gateway integration (Paymob, Apple Pay, Stripe), and Microsoft Azure deployment. Strong foundation in Object-Oriented Programming (OOP), SOLID principles, and design patterns, with end-to-end ownership across system design, database modeling, API engineering, responsive UI development, and production deployment.

CORE COMPETENCIES

Full-Stack Web Development | .NET 8 | ASP.NET Core Web API | C# | Angular | TypeScript | Clean Architecture | CQRS | RESTful API Design | Microservices Concepts | SignalR | Redis | SQL Server | Entity Framework Core | Dapper | Microsoft Azure | CI/CD | OAuth 2.0 | JWT | Role-Based Access Control (RBAC) | Payment Gateway Integration | Agile | Scrum | Object-Oriented Programming (OOP) | SOLID Principles | Design Patterns | Unit Testing

TECHNICAL SKILLS

Programming Languages: C#, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, Sass, T-SQL

Backend Development: .NET 8, ASP.NET Core Web API, ASP.NET MVC, Entity Framework Core (EF Core), Dapper, LINQ, AutoMapper, FluentValidation, API Versioning (Asp.Versioning), Swagger, OpenAPI, Background Services, Custom Middleware, Dependency Injection, JSON Web Tokens (JWT), OAuth 2.0

Frontend Development: Angular 17, Angular 18, Angular 20, Angular 21, Standalone Components, RxJS, Reactive Forms, Route Guards, HTTP Interceptors, Server-Side Rendering (SSR), Angular Material, Angular CDK, PrimeNG, Tailwind CSS, Bootstrap, Chart.js, ng2-charts, ngx-charts, ngx-translate, jsPDF, html2canvas, SweetAlert2

Databases and ORM: Microsoft SQL Server, T-SQL, Stored Procedures, Indexing, Query Optimization, Entity Framework Core Migrations (Code-First), EF Core Interceptors, Dapper (hybrid CQRS), Repository Pattern, Unit of Work Pattern, Generic Repository, ASP.NET Identity

Architecture and Design Patterns: Clean Architecture (multi-layer), Layered Architecture, N-Tier Architecture, Domain-Driven Design (DDD), CQRS, SOLID Principles, OOP, Repository and Unit of Work, Service Manager Pattern, Mediator Pattern, Factory Pattern, Singleton Pattern, Strategy Pattern, Dependency Injection (DI), Inversion of Control (IoC)

Real-Time, Caching, and Security: SignalR (multi-hub architecture), SignalR Redis Backplane, Redis, StackExchange.Redis, Distributed Caching, JWT with Refresh Tokens, OAuth 2.0 (Google, Facebook), One-Time Password (OTP) via SMS, BCrypt password hashing, HMAC-SHA512 webhook verification, RBAC, CORS, Rate Limiting, OWASP best practices

Integrations and Cloud: Paymob Payment Gateway, Apple Pay, Stripe, Firebase Admin SDK, Firebase Cloud Messaging (FCM), Twilio (SMS and OTP), EmailJS, Crystal Reports, Microsoft Azure App Service, Azure Blob Storage, Azure SQL Database, Vercel, GitHub Actions, CI/CD

Tools and Methodologies: Visual Studio 2022, Visual Studio Code, Git, GitHub, Postman, Swagger UI, SQL Server Management Studio (SSMS), JIRA, Figma, Agile, Scrum, Code Review, Unit Testing (Jasmine, Karma)

Languages: Arabic (Native), English (Intermediate, Professional Working Proficiency)

PROFESSIONAL EXPERIENCE

Senior Full-Stack Developer

BDC (Business Development Company), Cairo, Egypt | October 2025 - Present

Lead full-stack development of enterprise-grade web platforms using .NET 8, ASP.NET Core, Angular, and Microsoft Azure.

- Architect and deliver production-grade platforms using Clean Architecture (6 layers) and hybrid CQRS combining Entity Framework Core 8 and Dapper, while applying SOLID principles.
- Design and implement real-time communication systems with SignalR multi-hub architecture and Redis backplane for distributed scenarios across multiple server instances.
- Integrate payment gateways including Paymob Intention API and Apple Pay with HMAC-SHA512 webhook verification, achieving PCI compliance.
- Implement secure authentication using JWT with refresh tokens, OAuth 2.0 (Google and Facebook), OTP via Twilio, and Role-Based Access Control (RBAC).
- Develop Angular 20 and Angular 21 admin dashboards with 25+ feature modules, Angular Material, Chart.js, and PDF report generation using jsPDF and html2canvas.
- Configure Firebase Cloud Messaging (FCM) for push notifications, Azure Blob Storage for media assets, and Azure deployment pipelines using GitHub Actions.
- Lead technical decisions, mentor team members, conduct code reviews, and document architectural patterns to ensure long-term maintainability.

Frontend and Backend Developer

Solution Software | May 2021 - September 2025

Developed enterprise web applications for Saudi Arabian municipality clients across warehouse management, equipment maintenance, and real estate investment platforms.

- Architected scalable RESTful APIs using ASP.NET MVC and ASP.NET Core with Code-First Entity Framework, applying Clean Architecture and SOLID principles.
- Designed multi-screen user interface architectures (popup, flat, header-detail) to support complex enterprise data workflows.
- Integrated Crystal Reports for advanced reporting, analytics, and business intelligence dashboards.
- Implemented Role-Based Access Control (RBAC) with fine-grained permissions across multiple user roles.
- Collaborated within Agile and Scrum teams; conducted code reviews and contributed to technical decision-making.

Freelance Full-Stack Developer

Self-Employed | November 2020 - 2021

- Built end-to-end web platforms for diverse clients including online shopping, donate-medicine, and food delivery applications.
- Delivered Angular front-ends integrated with .NET Core back-ends, RESTful APIs, and SQL Server databases.
- Designed responsive user interfaces and managed the full deployment lifecycle from staging to production.

KEY PROJECTS

Voltyks - Electric Vehicle Charging Platform (v2.2.0)

Technologies: .NET 8, Clean Architecture (6 layers), Angular 20, Entity Framework Core, Dapper, SignalR, Redis, Microsoft Azure

Production-grade enterprise platform connecting electric vehicle owners with charging station providers; deployed on Microsoft Azure.

- Architected backend across 6 layers (API, Application, Core, Infrastructure, Persistence, Admin Dashboard) with hybrid CQRS combining EF Core 8 for writes and Dapper for high-performance reads.
- Designed real-time charging request system using SignalR with Redis backplane across three hubs (ChargingRequest, Notification, Process).
- Integrated Paymob Intention API with Apple Pay, card tokenization, saved cards, and HMAC-SHA512 webhook verification (PCI compliant).
- Implemented JWT authentication with refresh tokens, Google and Facebook OAuth 2.0, OTP via Twilio, and RBAC across 18 controllers.
- Built Angular 20 admin dashboard with 25+ feature modules, Angular Material, Chart.js, and PDF report generation using jsPDF and html2canvas.

Sekka API and Admin Panel - Smart Delivery Management Platform

Technologies: .NET 8, Clean Architecture, Angular 21, Entity Framework Core, FluentValidation, Redis, SignalR

Backend powering 579+ API endpoints, 83+ services, 82 database tables, 11 background services, and 4 SignalR hubs.

- Engineered .NET 8 Clean Architecture across 6 projects with API Versioning, FluentValidation, AutoMapper, and Redis distributed caching.
- Designed 4 SignalR hubs (OrderTracking, Chat, CashAlert, Notification) for real-time operations.
- Implemented 11 background services for scheduled jobs, settlements processing, RFM customer analysis, surge pricing, and analytics.
- Built 4 custom middleware and EF Core interceptors for audit trails and soft-delete.
- Developed Angular 21 admin panel with real-time monitoring dashboards using Chart.js and ng2-charts.

LogiFit - Fitness and Logistics Application

Built with Angular 18 and PrimeNG 17 with i18n (ngx-translate); visualized analytics via ngx-charts and generated PDF/Word reports using jsPDF, jsPDF-autotable, docx, and file-saver. *Tech: Angular 18, PrimeNG 17, Tailwind CSS, ngx-translate, ngx-charts, jsPDF, Vercel*

E-Commerce Platform - Full-Stack Solution

Angular 17 SSR front-end with ASP.NET Core + EF Core backend (Layered Architecture, Repository + Unit of Work); integrated Stripe, JWT with refresh tokens, and Swagger/OpenAPI. *Tech: Angular 17 SSR, ASP.NET Core Web API, EF Core, SQL Server, Stripe, JWT*

Social App - Angular 17 SSR Social Network

Social networking platform built with Angular 17 Server-Side Rendering, Express server, and Tailwind CSS; JWT authentication with jwt-decode. *Tech: Angular 17 SSR, Express, Tailwind CSS, Flowbite, JWT*

Maintenance Management System

Maintenance system for an automotive/equipment servicing company; multiple UI structures (popup, flat, header-detail), Crystal Reports, RBAC, and SOLID principles. *Tech: ASP.NET MVC, EF Code-First, Crystal Reports, SOLID, RBAC*

Personal Portfolio

Modern responsive bilingual (Arabic/English) portfolio with AOS animations and EmailJS contact form. *Tech: Angular 20, Tailwind CSS, AOS, EmailJS, Vercel*

SPEAKING ENGAGEMENTS

Guest Speaker - Career Bridge Workshop

Arab Open University (AOU), Egypt | May 2026

- Delivered a talk on the evolving role of developers in the AI era and how to leverage AI tools effectively without over-reliance.
- Invited by the Faculty of Computer Studies to share real-world insights and career guidance alongside industry experts.

EDUCATION

Bachelor of Computer Science

Higher Technological Institute, Egypt

Graduated: September 2021 | General Grade: Good | Graduation Project: Excellent

CERTIFICATIONS AND DIPLOMAS

- Frontend Developer Diploma - Route Academy: HTML5, CSS3, JavaScript, TypeScript, Angular, jQuery, Bootstrap, Tailwind CSS, Sass.
- Backend Developer Diploma - Route Academy: C#, OOP, Advanced C#, SQL Server, .NET Core.
- Full Stack Web Development Diploma - Russian Cultural Center: Frontend and Backend development with focus on .NET and ASP.NET MVC.
- Backend Course - egypt_is: Agile, Unified Modeling Language (UML), C#, OOP fundamentals, ASP.NET MVC Code-First.

LANGUAGES

- Arabic: Native
- English: Intermediate (Professional Working Proficiency)